

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers — the numbers and the contexts are the only things that differ. Work every problem with no notes, then check yourself against the key. **80 points total.**

Part A — Vocabulary (10 pts)

Write the letter of the best description next to each term.

- | | |
|---------------------------------|--|
| 1. Function _____ | (A) The set of every output the rule actually produces. |
| 2. Domain _____ | (B) Feeding the output of one rule into another: $f(g(x))$. |
| 3. Range _____ | (C) A rule that pairs each input with exactly one output. |
| 4. Zero of a function _____ | (D) The rule that gives back the question: if $f(a) = b$, it sends b to a . |
| 5. Concave up _____ | (E) An input whose output is 0. |
| 6. Average rate of change _____ | (F) The simplest member of a family, before any shift or stretch. |
| 7. Parent function _____ | (G) The set of every input the rule is allowed to take. |
| 8. Composition _____ | (H) The one steady rate that would give the same change in output over the same run. |
| 9. One-to-one _____ | (I) Bending upward — the rate of change is increasing. |
| 10. Inverse function _____ | (J) No two different inputs share an output. |

Part B — Multiple Choice (12 pts)

Circle the best answer, then write its letter in the blank at the right.

- The statement $f(3) = 7$ tells you that
 (A) $f(7) = 3$ (B) 7 is an input of f (C) $(3, 7)$ is on the graph of f (D) f is increasing at $x = 3$ _____
- On an interval where f is decreasing and concave up, the outputs are
 (A) falling faster and faster (B) rising more and more slowly (C) changing at a constant rate (D) falling, but more and more slowly _____
- A table of f has equally spaced inputs and **constant second** differences. The best model is
 (A) linear (B) quadratic (C) exponential (D) none of these _____
- The graph of $g(x) = f(x + 3)$ is the graph of f shifted
 (A) 3 left (B) 3 right (C) 3 up (D) 3 down _____
- If $f(x) = x^2$ and $g(x) = x - 1$, then $(f \circ g)(4)$ equals
 (A) 15 (B) 16 (C) 9 (D) 3 _____
- A function has an inverse that is also a function exactly when the function is _____

Part C — Short Answer & Computation (40 pts, 5 each)

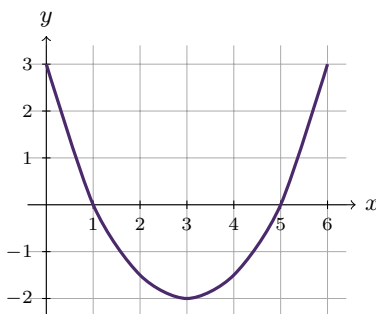
Show your work in the space provided.

1. Let $f(x) = 2x^2 - 5$.

(a) $f(3) =$ _____ (b) $f(-2) =$ _____

(c) Solve $f(x) = 3$. There are two answers: _____

(d) Which of (a)–(c) is a *solve* question rather than an *evaluate* question? _____

2. The graph of $y = f(x)$ is shown below.

(a) On what interval is f decreasing? _____

(b) Concave up or concave down on the whole picture? _____

(c) List the zeros of f : _____

(d) $f(3) =$ _____ (e) Range on the window shown: _____

3. A tank is draining. Its volume V (gallons) after t minutes:

t (min)	0	4	8	12
V (gal)	180	148	124	108

(a) Average rate of change of V from $t = 0$ to $t = 8$: _____, with units _____.

(b) In one sentence, what does that number mean for the tank?

(c) Is the tank draining faster or slower as time goes on? _____ Justify with the numbers.

4. The table below has equally spaced inputs. Fill in the difference rows, then name the function type.

x	0	1	2	3	4
$g(x)$	2	5	12	23	38
1st differences	—	_____	_____	_____	_____
2nd differences	—	—	_____	_____	_____

Function type: _____ Because _____

5. The parent function is $f(x) = \sqrt{x}$, with domain $[0, \infty)$ and range $[0, \infty)$. Let $g(x) = -2f(x - 1) + 4$.

- (a) Write g in terms of x : $g(x) =$ _____
- (b) The change *inside* the parentheses moves the graph _____
- (c) The changes *outside*, in order: _____
- (d) Domain of g : _____ Range of g : _____
- (e) The parent's landmark point $(0, 0)$ lands at _____ on g .

6. Two one-to-one functions are given by the table below.

x	0	1	2	3	4
$f(x)$	5	2	7	1	3
$g(x)$	3	0	4	2	1

- (a) The handoff in $f(g(1))$ is _____, so $f(g(1)) =$ _____
- (b) $g(f(3)) =$ _____
- (c) $f^{-1}(7) =$ _____ because $f(\text{_____}) = 7$.
- (d) How can you tell from the table that f is one-to-one?
-

(e) Why does $g(f(0))$ have no value from this table?

7. Let $f(x) = 3x + 1$ and $g(x) = x^2 - 2$.

- (a) $f(g(2)) =$ _____ (b) $g(f(2)) =$ _____

(c) $(f \circ g)(x) =$ _____

(d) What do (a) and (b) show about the order of a composition?

8. Let $f(x) = \frac{x-5}{2}$.

(a) Find f^{-1} by swapping and solving: $f^{-1}(x) =$ _____

(b) Verify: $f(f^{-1}(x)) =$ _____

(c) $f(11) = 3$, so $f^{-1}(3) =$ _____

Part D — Extended Response (18 pts, 9 each)

Explain your reasoning in full sentences.

1. A rectangular dog run is built against a barn wall, so only **three** sides need fencing. The builder has 60 feet of fence. Let x be the length of each of the two sides perpendicular to the barn.

(a) Which quantity is the *question* the model answers — its input? _____

(b) The side parallel to the barn has length _____, so $A(x) =$ _____

(c) What type of function is A , and how do you know?

(d) Domain the context forces: _____ $< x <$ _____ Why?

(e) Average rate of change of A from $x = 5$ to $x = 10$: _____, with units _____.

(f) State one assumption this model rests on.

2. Let $f(x) = x^2$ with domain all real numbers.

(a) Explain why f has no inverse function on that domain. Give a counterexample pair of inputs.

(b) Give a domain restriction that fixes the problem: _____ Then $f^{-1}(x) =$ _____

(c) Domain of that inverse: _____ Range: _____

(d) Explain how the graph of a function and the graph of its inverse are related, and why.
